

Hyeonjae Kim

Email: hyeonjae.amo@gmail.com
 hyeonjae@add.re.kr
Website: hyeonjae.com

Research Officer at the Agency for Defense Development

Education

Kyungpook National University (KNU)	Daegu, Republic of Korea
Bachelor of Science in Physics	Mar. 2020 – Feb. 2024
Overall GPA: 4.11/4.3 Major GPA (71 credits): 4.19/4.3	

Research Experiences

Agency for Defense Development (ADD)	Daejeon, Republic of Korea
Research Officer (Supervisors: Dr. Sin Hyuk Yim, Dr. Sangkyung Lee)	Jun. 2024 – May 2027 (expected)
Research in atomic physics and quantum sensing	
• Designed and implemented an automated absorption spectroscopy system to characterize rubidium vapor cell properties • Built and optimized a zero-field (SERF) optically pumped magnetometer for magnetic noise measurements • Conducted magnetic anomaly detection using a drone-mounted all-optical pulsed atomic magnetometer and characterized the resulting anomaly signal	

Novel Applied Nano Optics Lab	Daegu, Republic of Korea
Research Intern (Advisor: Prof. Junyeob Yeo)	Jun. 2021 – Jun. 2022
Project: Fabrication of a flexible photo-electrochemical cell using laser	

High Energy Physics Lab (Moon Lab)	Daegu, Republic of Korea
Research Intern (Advisor: Prof. Chang-Sung Moon)	Jun. 2020 – Feb. 2021
Developed momentum reconstruction algorithms for silicon detector simulations	

Publications

Magnetic anomaly detection of an undersea tunnel using a drone-mounted all optical pulsed atomic magnetometer
Manuscript in preparation
H. Kim, S. H. Yim*, S. Lee, E. -S. Bang, S. R. Yeo, Y. J. Kim, T. Jeong, and Y. Lim

Novel sealing of glass-blown atomic gas cells using resistive heating wires
Manuscript under review
S. H. Hong, Y. Lim*, S. H. Yim, S. Lee, H. Kim, T. Jeong, and J. B. Nam

Aging test of an atomic vapor cell with Al₂O₃ wall coating on cubic glass
Applied Optics **64**, 7932-7937 (2025)
H. Kim, T. Jeong, S. H. Hong, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim*

Selected Conference Presentations

Probing magnetic noise from portable sensor components using a zero-field optically pumped magnetometer
Poster presentation, APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting, 2026

Experimental systems for spectroscopy and zero-field OPM with atomic vapor cells
Oral presentation (Invited), KISTI-SNU Joint workshop (QNRC Joint Workshop), 2026

Towards the development of a SERF magnetometer

Poster presentation, Korean Physical Society (KPS) Fall Meeting, 2025

Lifetime extension of rubidium vapor cells by Al₂O₃ coating

Poster presentation, APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting, 2025

Linear absorption spectroscopy in rubidium vapor cells: Applications in buffer gas measurement and lifetime estimation

Poster presentation, Korean Physical Society (KPS) Spring Meeting, 2025

Honors and Awards

Physics Alumni Association Award and Scholarship (Outstanding Graduate)	Feb. 2024
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Department of Physics Alumni Association, Kyungpook National University

2nd Prize, 2023 MiliTECH Challenge	Dec. 2023
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Hosted by KAIST; Awarded by the Agency for Defense Development

2nd Prize, 2022 MiliTECH Challenge	Dec. 2022
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Hosted by KAIST; Awarded by the Korea Military Academy (KMA)

3rd Prize of Academic Conference	Dec. 2021
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College of Natural Sciences, Kyungpook National University

Dean's List: 3 semesters (2021-1, 2021-2, 2022-2)	Nov. 2021, Apr. 2022, Apr. 2023
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College of Natural Sciences, Kyungpook National University

Grants and Scholarships

National Science and Technology Scholarship	Mar. 2022 – Feb. 2024
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Korea Student Aid Foundation (KOSAF)

Academic Merit Scholarship (Hyoseok Scholarship)	Apr. 2022
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Kyungpook National University Alumni Association

Academic Excellence Scholarship	Aug. 2020, Feb. 2021, Aug. 2021
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Kyungpook National University

Other Research Outputs

Buffer Gas Pressure Estimation for OPMs	Dec. 2024
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Registered software; Korea Copyright Commission (C-2024-047156)

Rubidium Vapor Cell Lifetime Monitoring and Analysis Tool	Dec. 2024
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Registered software; Korea Copyright Commission (C-2024-054414)

Additional Experiences

Selected Participant, KFAS Study Abroad Workshop	Jul. 2026 – Jul. 2027
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Natural Sciences; Korea Foundation for Advanced Studies (KFAS)

Research Officer for National Defense (ROND)	Jun. 2024 – May 2027
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First Lieutenant in the Republic of Korea Air Force

Selected as one of 20 research officers nationwide for STEM-based national defense research

Teaching Assistant, Classical Mechanics II (PHYS212)	Sep. 2023 – Dec. 2023
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Teaching Assistant, Electromagnetism II (PHYS312)	Mar. 2023 – Jun. 2023
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Skills

Programming: Python, Wolfram Language (Mathematica)

Software: LabVIEW, SolidWorks, COMSOL Multiphysics